Values variables and types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Values variables and types before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Values variables and types trace and record the first step you can explain without help.

Use W2: Trace a saved **value** for **Value**s **variable**s and **type**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Values variables and types explanation against the supplied facts before discussing it.

Use the worked case for **Value**s **variable**s and **type**s. Proposed explanation: “Digits always mean a number”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Trace values, types and reassignment

Predict both **output** lines and explain why estimate stays unchanged when count is reassigned. Repeat the trace with initial count=0. Explain why note is text even though its printed characters look like a number.

```
count = 4
note = "4"
estimate = count / 2
fits = count <= 5
count = count + 3
print(count, note, estimate, fits)
print(type(count).__name__, type(note).__name__, type(estimate).__name__,
type(fits).__name__)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Identify the value and its type

> Identify the value and its type. Apply the supplied Values variables and types facts and keep your reasoning.

Classify 12, 1.5, "12", False and "False". Explain why the last two are different.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace a saved value

> Trace a saved value. Apply the supplied Values variables and types facts and keep your reasoning.

Trace the five-line count and saved program. Record both printed **output**s, then predict the result of assigning saved = count after the change. Check with a run or label a paper trace.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Diagnose the operation

Diagnose the operation. Apply the supplied Values variables and types facts and keep your reasoning.

Predict 6 + 3, "6" + "3" and "6" * 3. Explain what happens for "6" + 3. Which expression counts six groups of three?

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Trace a new pair of names independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Trace a new pair of names independently. Apply

> the supplied Values variables and types facts and keep your reasoning.

A fresh script runs seats = 12; booked = seats; seats = seats - 5; print(seats); print(booked), with each statement on a separate line. Predict both **output**s. Then add booked = seats + 1 and print(booked).

Explain the evaluation order and choose a **type** for a seat label "07".

> **My reasoning and evidence**

WORKBOOK · 9 OF 15

W5 Separate type from validity

> Separate type from validity. Apply the supplied Values variables and types facts and keep your reasoning.

Give a suitable **type** for a counter count, a length of 2.5 metres, a shelf ID with a leading zero, and a yes-or-no availability flag. Give a count **value** with the right **type** that should still be rejected.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Values variables and types to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled **value**s **variable**s and **type**s **evidence** sheet to Python readiness portfolio. Show how you can distinguish integer, floating-point, text and boolean **value**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Values variables and types

> A number and its printed-looking text. A program stores a numeric badge count and a string that looks like that count. Explain the labelled step before checking the changed condition.

A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count.

Python 3: count=8; tag="8"; print(count+2); print(tag+"2").

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Values variables and types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Inspect types

count is an integer; tag is a string.

2. Apply addition

Integer addition gives 10; string concatenation gives the text 82.

3. Predict a mixed operation

count+tag raises TypeError rather than silently converting tag.

CHANGE ONE THING

Change count to 10, leaving tag unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? count+tag raises **Type**Error rather than silently converting tag.

1. Predict a mixed operation

2. Inspect **type**s

3. Apply addition

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Predict a mixed operation count+tag raises **Type**Error rather than silently converting tag. **Value**s carry **type**s, not merely appearances.

The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change count to 10, leaving tag unchanged. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The two valid **output**s become 12 and 82.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The two valid **output**s become

12 and 82. Only the integer **variable** changes; the **string** still contains the character 8.

Repair a missing building block

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

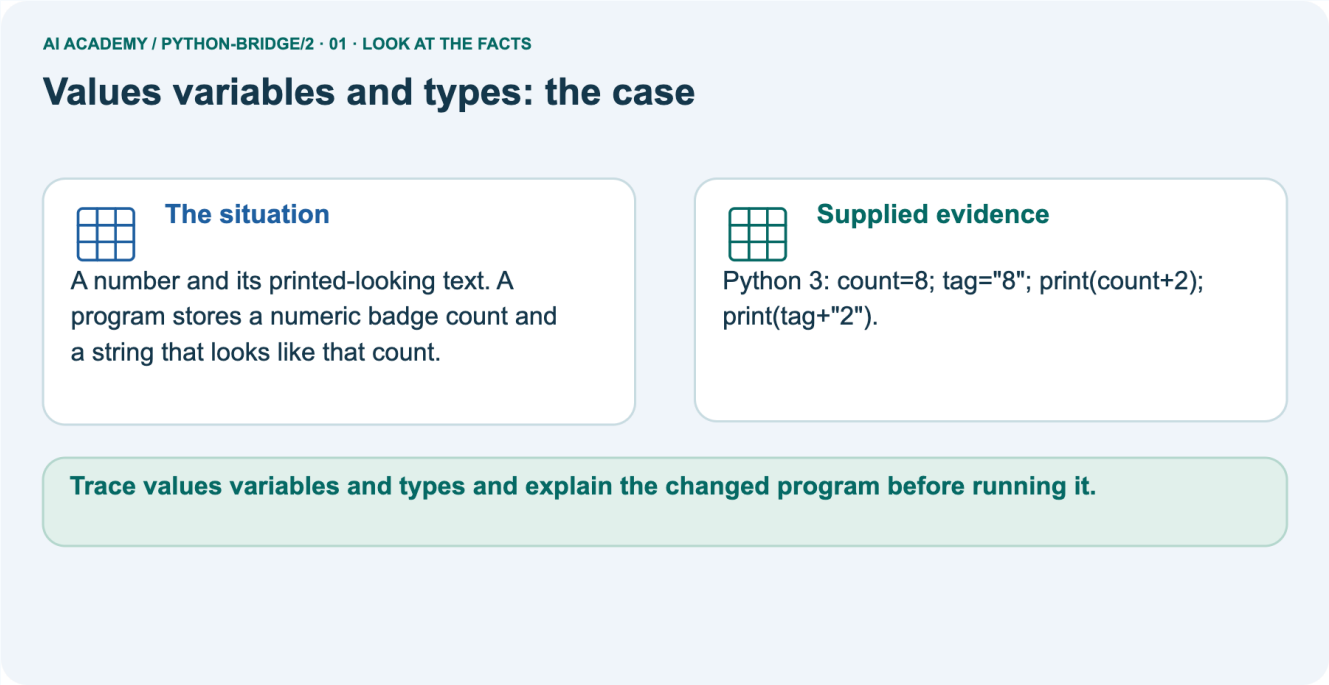
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

Explain each Values variables and types picture from its labels, then record a relevant limit.

01. Values variables and types: the case



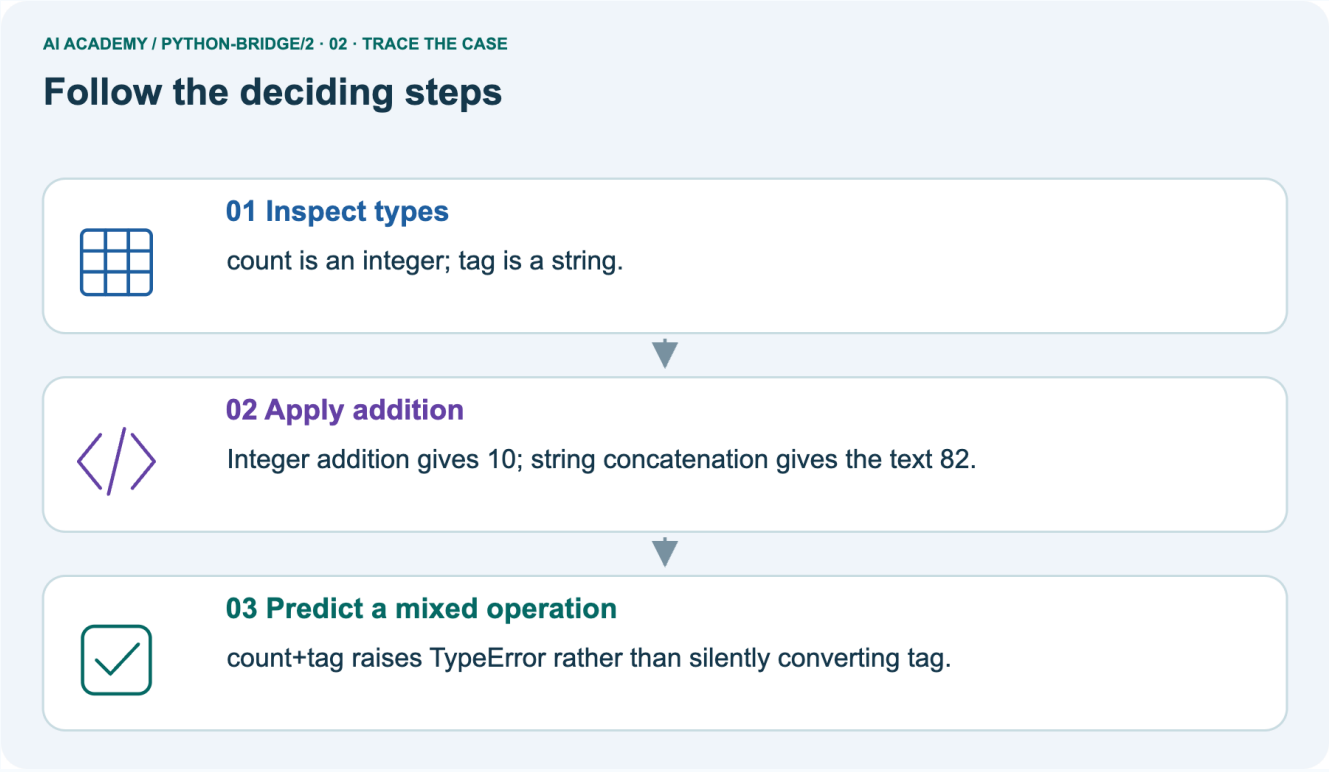
Follow the labelled picture with the matching explanation. Open the vector diagram

A number and its printed-looking text. A program stores a numeric badge count and a **string** that looks like that count. Supplied facts: Python 3: count=8; tag="8"; print(count+2); print(tag+"2").

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect **type**s: count is an integer; tag is a **string**. → Apply addition: Integer addition gives 10; **string** concatenation gives the text 82. → Predict a mixed operation: count+tag raises **Type**Error rather than silently converting tag.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A saved value is not a live link

AI ACADEMY / PYTHON-BRIDGE/2 · 03 · INSPECT THE LESSON MODEL

A saved value is not a live link

Instruction	count now	saved now
count = 8	8	Not assigned
saved = count	8	8
count = count + 2	10	8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

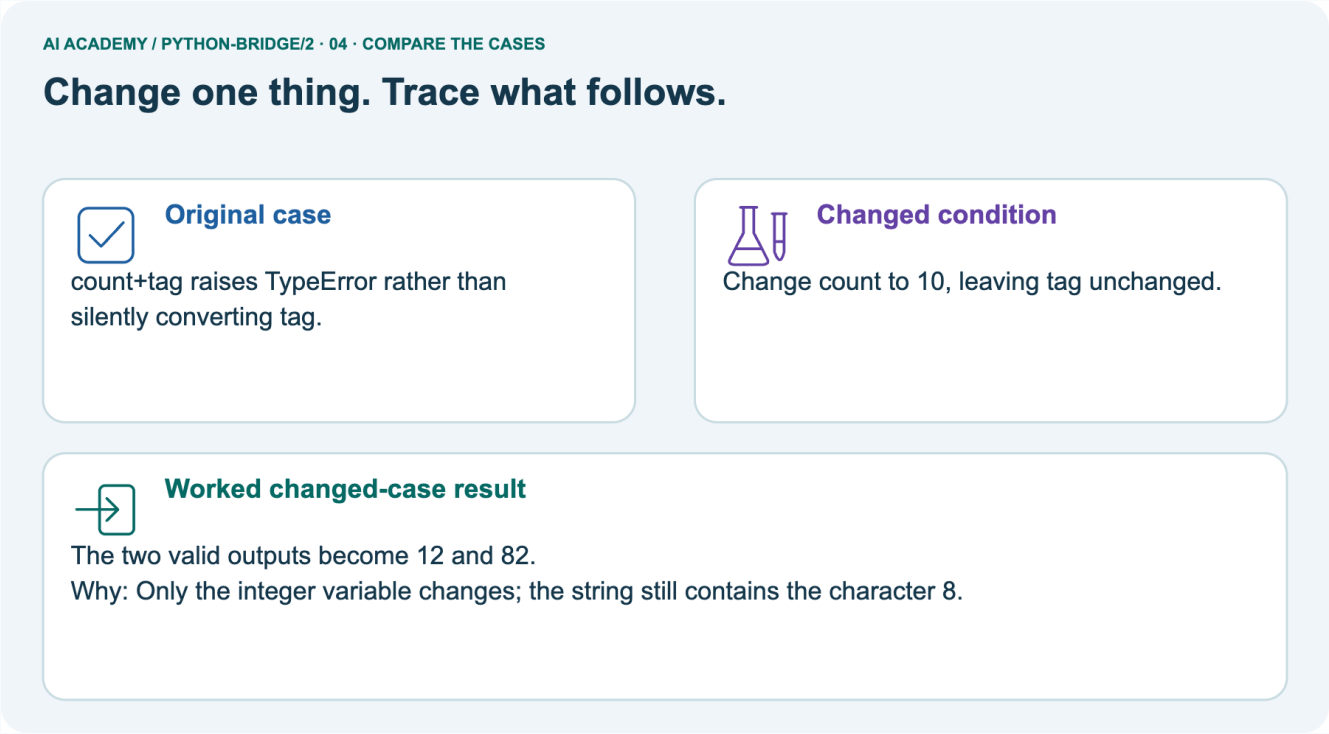
Follow the labelled picture with the matching explanation. Open the vector diagram

A saved **value** is not a live link. count = 8 | 8 | Not assigned. saved = count | 8 | 8. count = count + 2 | 10 | 8

Read the labels and **value**s. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

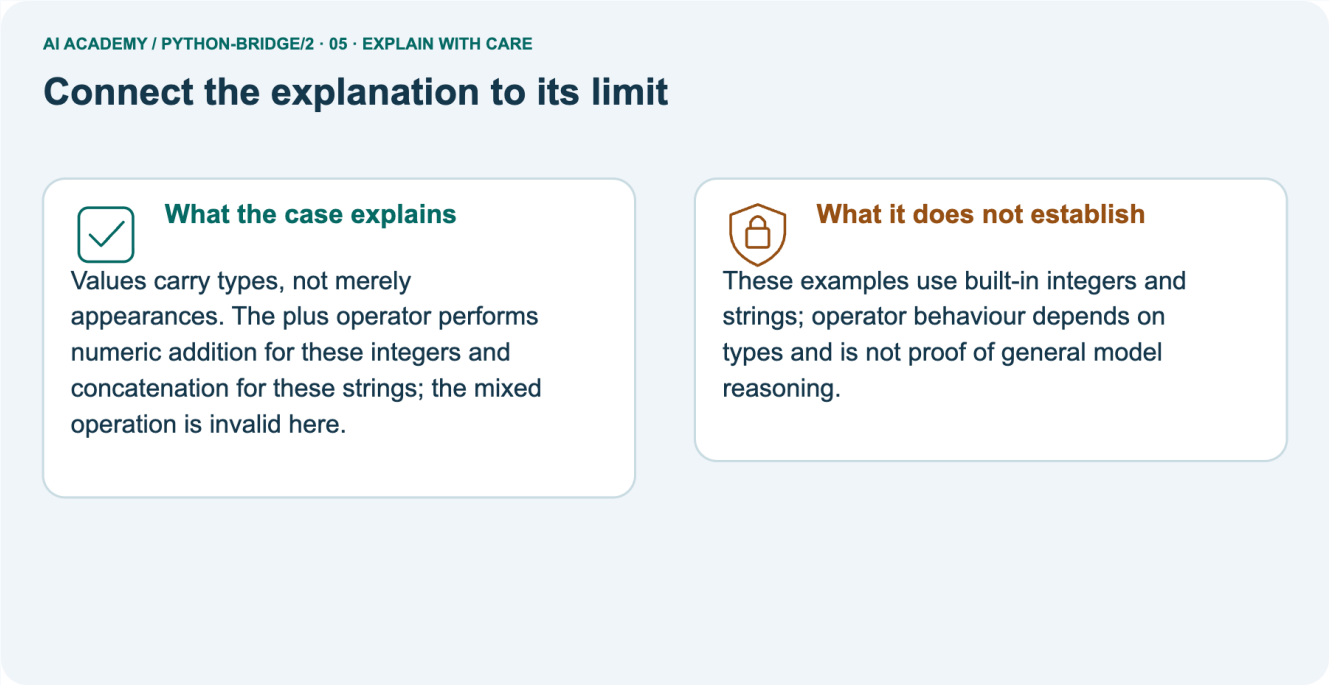
Original case: count+tag raises **Type**Error rather than silently converting tag. Changed condition: Change count to 10, leaving tag unchanged. Expected result: The two valid **output**s become 12 and 82.

Reason: Only the integer **variable** changes; the **string** still contains the character 8.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

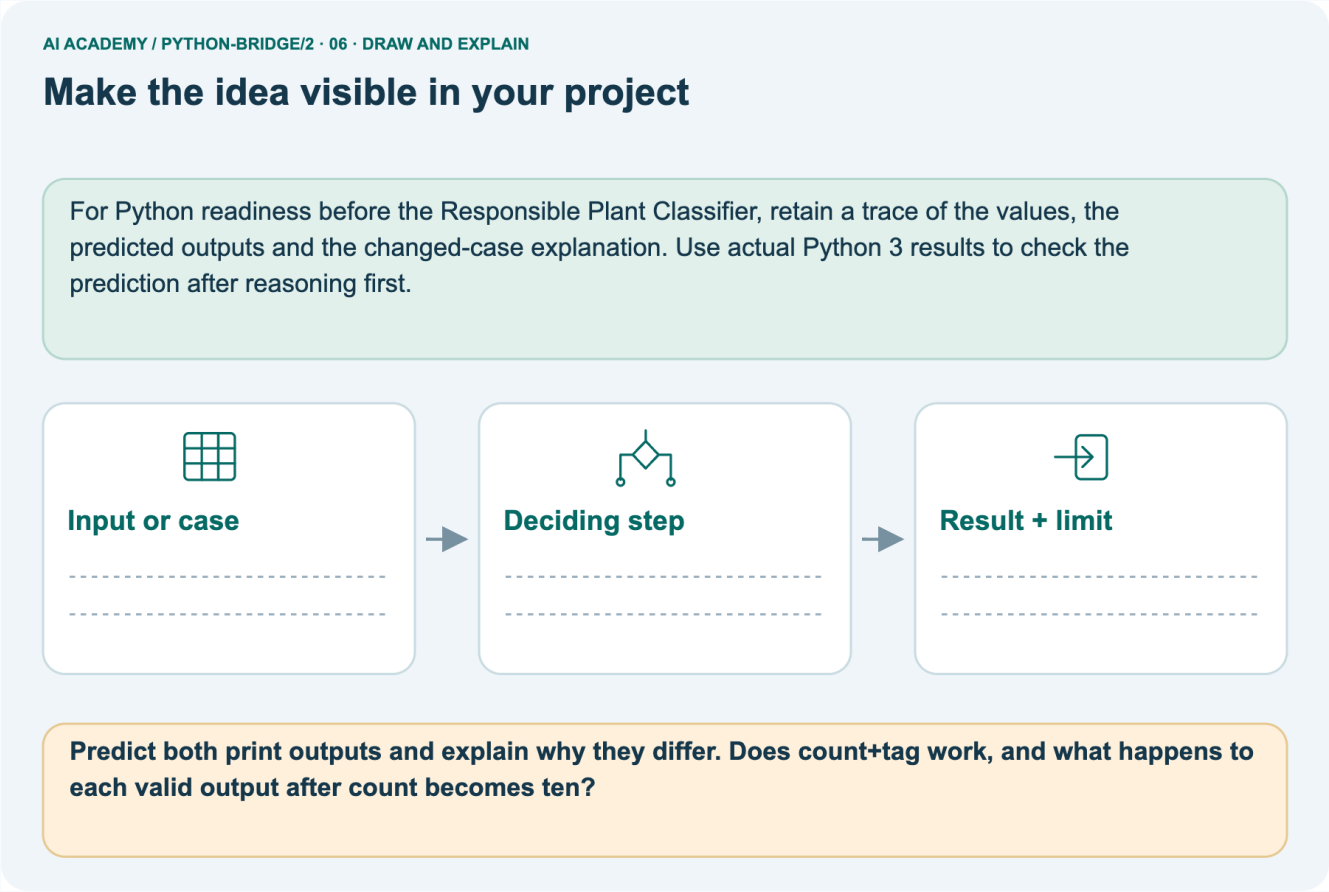
Values carry **type**s, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these **string**s; the mixed operation is invalid here.

Limit: These examples use built-in integers and **string**s; operator behaviour depends on **type**s and is not proof of general model reasoning.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the **value**s, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict both print **output**s and explain why they differ. Does count+tag work, and what happens to each valid **output** after count becomes ten?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.